

Concentric Tube Continuum Robots (CTCRs) are an emerging technology in medical robotics used for tool traversal in minimally invasive surgeries. The central control mechanism of CTCRs consists of multiple precurved telescoping and rotating tubes made of superelastic nitinol whose elastic interactions produce controlled bending of the distal end-effector, allowing for a dexterous workspace. Compared to the more common Tendon Driven Continuum Robots (TDCRs), CTCRs are more amenable to miniaturization and do not require external tendons, enabling more complex navigation. However, the workspace is inherently limited by the combination of geometries in the constituent tubes. That leads to a bottleneck where the versatility of the robot depends on rapid fabrication of unique tube curvatures. The temperature-sensitive shape-setting behavior of nitinol tubes makes this a highly variable and laborious process.

The goal of this study is to [1] build a CTCR system and evaluate its accuracy and effectiveness and [2] improve the repeatability and flexibility of the heat-setting process for nitinol tubes to standardize fabrication. We created a prototype based on an open-source design and are currently developing an embedded system to shape-set the tubing. We are using a common approach where the tubes are placed in a calcium silicate fixture and heated through Joule heating using a custom PCB.